5.3: The Other Trigonometric Functions

Definition.

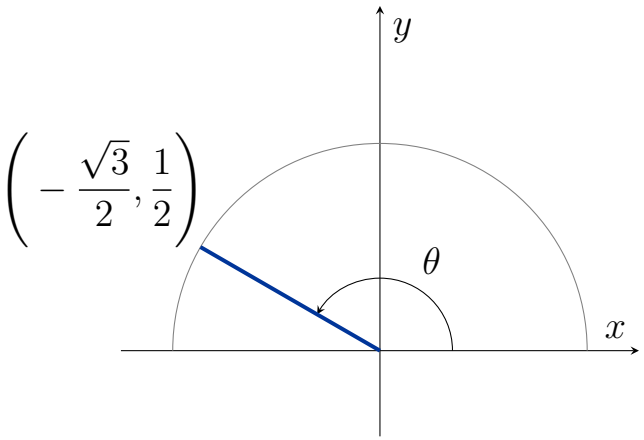
$$\sec(\theta) = \frac{1}{\cos(\theta)} \qquad \tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)} \qquad \cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

Example. Simplify the following:

$$\frac{\sec(\theta)}{\tan(\theta)} \qquad \tan(\theta)\cot(\theta)$$

Example. Consider the point $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ on the unit circle. Find the value of the six trigonometric functions:



$\cos(\theta)$

$\sin(\theta)$

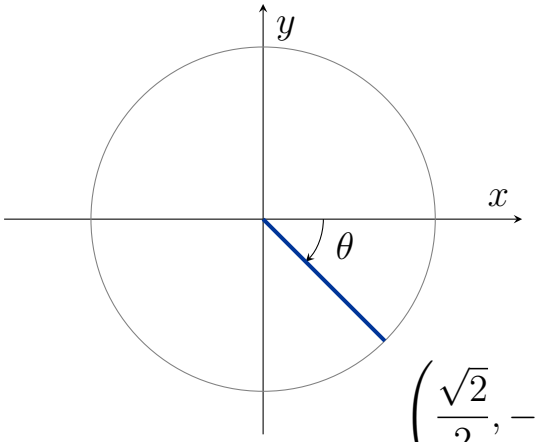
$\sec(\theta)$

$\csc(\theta)$

$\tan(\theta)$

$\cot(\theta)$

Example. Consider the point $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ on the unit circle. Find the value of the six trigonometric functions:



$\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

$\cos(\theta)$

$\sin(\theta)$

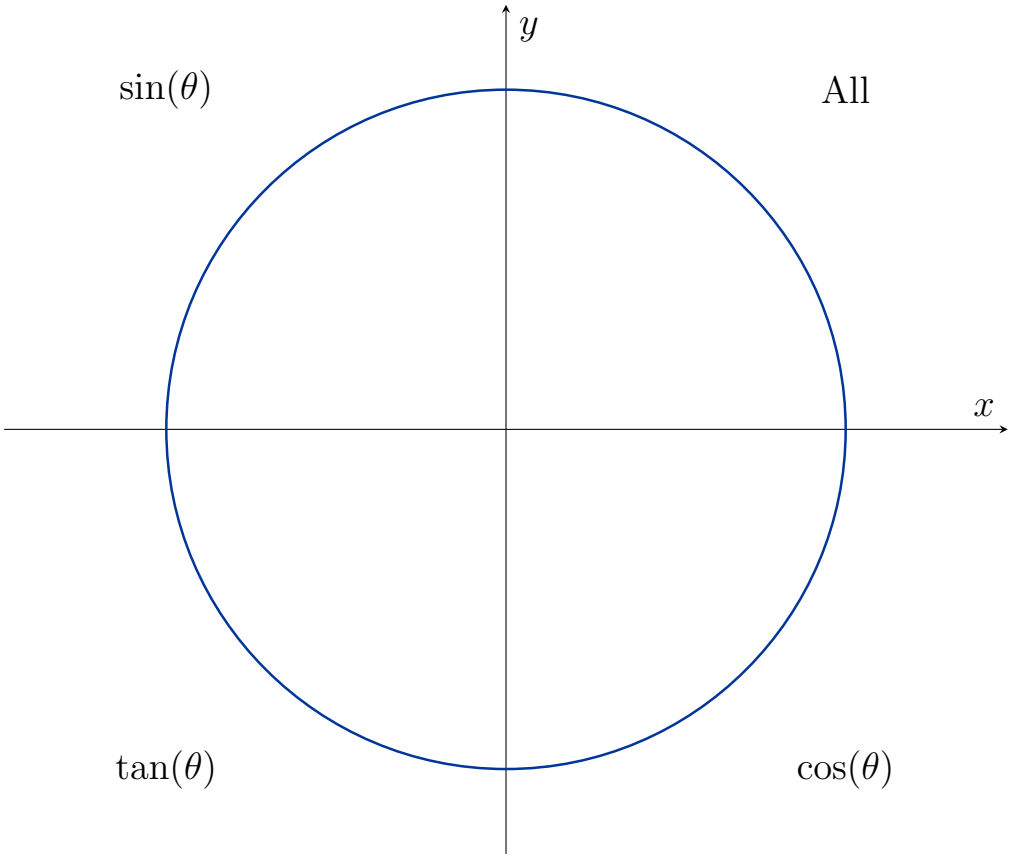
$\sec(\theta)$

$\csc(\theta)$

$\tan(\theta)$

$\cot(\theta)$

Note: The sign of each trig. function is determined by the quadrant



Example. Find the value of the six trigonometric functions when

$\theta = \dfrac{\pi}{6}$ $\theta = \dfrac{\pi}{3}$

$\theta = -\dfrac{5\pi}{6}$ $\theta = -\dfrac{7\pi}{4}$

Recall that $\cos(\theta)$ is even, and $\sin(\theta)$ is odd:

$$\cos(\theta) = \cos(-\theta) \qquad -\sin(\theta) = \sin(-\theta)$$

Thus,

$$\sec(\theta) = \sec(-\theta) \qquad -\csc(\theta) = \csc(-\theta)$$

$$-\tan(\theta) = \tan(-\theta) \qquad -\cot(\theta) = \cot(-\theta)$$

Example. If $\sec(\theta) = 2$, what is $\sec(-\theta)$?

Example. If $\cot(\theta) = \sqrt{3}$, what is $\cot(-\theta)$?

Alternate forms of the Pythagorean Theorem
Recall the Pythagorean Identity:

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

We now define alternate forms:

$$1 + \tan^2(\theta) = \sec^2(\theta)$$
$$\cot^2(\theta) + 1 = \csc^2(\theta)$$

Example. If $\tan(\theta) = -\sqrt{3}$, and $\pi < \theta < 2\pi$, find $\sec(\theta)$.

Example. If $\csc(\theta) = -\dfrac{2\sqrt{3}}{3}$, and $\pi < \theta < \dfrac{3\pi}{2}$, find $\cot(\theta)$.

Definition. (Period of a Function)
The **period** P of a repeating function f is the number representing the interval such that $f(x+P) = f(x)$ for any value of x.

The period for cosine, sine, secant, and cosecant is 2π.

The period for tangent and cotangent is π.

Example. If $\sin(\theta) = \frac{1}{5}$, find $\sin(\theta + 2\pi)$ and $\sin(\theta - 2\pi)$.

Example. If $\tan(\theta) = \frac{\sqrt{3}}{3}$, find $\tan(\theta + \pi)$ and $\cot(\theta + \pi)$.

Example. Find the value of all six trigonometric functions and identify which quadrant θ is in:

$$\cos(\theta) = \frac{5}{13} \text{ and } \sin(\theta) = -\frac{12}{13}$$

$$\csc(\theta) = -\sqrt{2} \text{ and } \tan(\theta) = -1$$